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Where Minds Pollinate

Class 10 Biology

Chapter 5

Short Questions Notes

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Chapter 5: Nervous System

Short Answer Questions

1. What is the nervous system?

The nervous system is the main control, coordination, and communication network of the human body. It consists of the brain, spinal cord, and nerves. It receives information from the surroundings through sense organs, interprets it, and sends commands to different body parts to respond properly. It helps us think, move, feel, and react to stimuli quickly.

2. What is the role of the nervous system?

The nervous system plays a vital role in maintaining coordination between different organs of the body. It controls voluntary actions like walking and speaking, as well as involuntary actions like heartbeat and breathing. It allows the body to sense changes in the environment and respond appropriately to maintain internal balance (homeostasis).

3. Differentiate between the central nervous system and the peripheral nervous system.

Central Nervous System (CNS)	Peripheral Nervous System (PNS)
Consists of the brain and spinal cord.	Consists of all nerves outside the brain and spinal cord.
Controls and coordinates all body functions.	Connects the CNS with limbs and organs.
Acts as the command centre of the body.	Acts as a communication line between the CNS and the rest of the body.

4. Name the parts of the human brain.

The human brain has three main parts:

1. **Cerebrum** – the largest part; controls thinking, memory, learning, and voluntary actions.
2. **Cerebellum** – controls balance, posture, and coordination of movements.
3. **Medulla oblongata** (brainstem) – controls involuntary actions such as heartbeat, breathing, and digestion.

5. Write two functions of the cerebellum.

1. The cerebellum maintains the body's balance and posture during movement.
2. It coordinates muscle actions so that movements are smooth and precise, such as walking or writing.

6. How are the brain and spinal cord protected?

The brain is protected by the **skull**, and the spinal cord is protected by the **vertebral column**. Both are covered by three protective membranes called **meninges**. Additionally, a clear fluid known as **cerebrospinal fluid (CSF)** surrounds them, providing cushioning and protection against shocks or injuries.

7. Describe the peripheral nervous system.

The peripheral nervous system (PNS) consists of all the nerves that branch out from the brain and spinal cord. It includes **12 pairs of cranial nerves** and **31 pairs of spinal nerves**. The PNS connects the central nervous system to limbs and organs, ensuring that information travels back and forth between the brain and body parts.

8. What are the types of neurons? What are their functions?

There are three main types of neurons:

1. **Sensory neurons** – carry information from sense organs (like skin, eyes, ears) to the brain or spinal cord.
2. **Motor neurons** – carry commands from the brain or spinal cord to muscles or glands for response.
3. **Interneurons** – connect sensory and motor neurons within the central nervous system and help in processing information.

9. Define stimulus with examples.

A **stimulus** is any change in the environment that can cause a response in an organism.

Example: When a person touches a hot object, the heat acts as a stimulus, and the quick withdrawal of the hand is the response.

10. Define and sketch a synapse.

A **synapse** is the tiny gap or junction between two neurons where a nerve impulse passes from one neuron to another. The transmission across a synapse occurs through chemicals called **neurotransmitters**.

(Diagram can show two neurons with synaptic gap and neurotransmitters.)

11. How does an impulse cross a synapse?

When an electrical nerve impulse reaches the end of a neuron, it triggers the release of neurotransmitters. These chemicals move across the synaptic gap and bind to the receptors on the next neuron, creating a new electrical impulse. This process continues until the signal reaches its destination.

12. What action or change do neurotransmitters transmit to the target cell?

Neurotransmitters cause the target cell (another neuron, muscle, or gland cell) to either **activate** or **inhibit** a specific response. For example, in muscles, they can cause contraction, while in glands, they can trigger the secretion of substances.

13. What happens to neurotransmitters after they deliver their message?

After transmitting the impulse, neurotransmitters are either broken down by enzymes, reabsorbed by the original neuron, or diffused away from the synapse to stop continuous stimulation.

14. How many different types of neurotransmitters are there?

There are many types of neurotransmitters, but some common ones are:

1. **Acetylcholine** – helps transmit signals to muscles.
2. **Dopamine** – controls movement and emotions.
3. **Serotonin** – affects mood and sleep.
4. **Adrenaline** – prepares the body for 'fight or flight' response.

15. What is a reflex action?

A reflex action is an automatic and quick response to a stimulus that does not involve thinking or the brain directly. It helps protect the body from harm.

Example: Pulling your hand away after touching something hot.

16. Name the endocrine glands of man.

The main endocrine glands are:

1. Pituitary gland
2. Thyroid gland

3. Parathyroid glands
4. Adrenal glands
5. Pancreas
6. Testes (in males)
7. Ovaries (in females)

17. Name the hormones secreted by each endocrine gland.

Gland	Hormone Secreted	Main Function
Pituitary gland	Growth hormone	Controls growth and other glands
Thyroid gland	Thyroxine	Controls metabolism
Parathyroid glands	Parathormone	Regulates calcium level
Adrenal glands	Adrenaline	Prepares body for emergencies
Pancreas	Insulin	Controls blood sugar level
Testes	Testosterone	Controls male reproductive functions
Ovaries	Estrogen and Progesterone	Controls the female reproductive cycle

18. Differentiate between given terms:

Terms	Difference
Cerebrum and Cerebellum	The cerebrum controls thinking, memory, and voluntary actions; the Cerebellum controls balance and muscle coordination.
Sensory and Motor Neuron	Sensory neurons carry impulses to the CNS. Motor neurons carry impulses from the CNS to muscles/glands.
Voluntary & Involuntary Action	Voluntary actions are under conscious control (walking); Involuntary actions occur automatically (breathing).
Nerve Impulse and Hormonal Transmission	A nerve impulse is an electrical and fast process; hormonal transmission is a chemical and slower process.
Endocrine and Exocrine Glands	Endocrine glands release hormones into the blood; Exocrine glands release secretions through ducts (like sweat glands).

Extensive Answer Questions

1. Describe the nervous system and its role in humans.

The nervous system is the body's master control and communication network. It is made up of the **central nervous system (CNS)** and the **peripheral nervous system (PNS)**.

The CNS includes the brain and spinal cord, which process information and send instructions. The PNS connects the CNS to all body parts through nerves.

Its main roles are:

- To receive information from sense organs.
- To interpret and process this information.
- To send appropriate responses to muscles and glands.

Through these actions, the nervous system helps maintain coordination, control voluntary and involuntary activities, and maintain balance in the body.

2. Draw a vertical section of the human brain and label it. Write the functions of each part.

(Diagram should show: Cerebrum, Cerebellum, Medulla Oblongata, and Spinal Cord)

Functions:

- **Cerebrum:** Controls thinking, memory, emotions, and voluntary movements.
- **Cerebellum:** Maintains balance and coordinates muscle actions.
- **Medulla oblongata:** Controls involuntary actions like heartbeat, breathing, and digestion.
- **Spinal cord:** Transmits messages between brain and rest of the body.

3. Draw and label cross section of human spinal cord and write its structure and function.

The spinal cord is a long, tube-like part of the nervous system that starts from the medulla oblongata in the brain and runs down through the vertebral column (backbone). It is protected by the bony vertebrae and works as a link between the brain and the rest of the body. It also controls reflex actions such as moving your hand away from something hot.

Structure:

When we look at the cross-section (cut section) of the spinal cord, we can see two main parts:

1. Grey Matter:

- It is in the center and looks like the letter “H” or a butterfly shape.
- It is made up of nerve cell bodies.
- The upper parts of the “H” are called dorsal (posterior) horns and the lower parts are called ventral (anterior) horns.
- The middle part has a small hole called the central canal, which is filled with cerebrospinal fluid (CSF).

2. White Matter:

- It is found around the gray matter.
- It looks white because it has myelinated nerve fibers (axons) that carry messages up and down the spinal cord.
- These nerve fibers make two types of tracts —
 - Ascending tracts: carry messages from body to brain (sensory).
 - Descending tracts: carry messages from brain to body (motor).

Other parts include:

- Dorsal root: brings sensory messages from sense organs to the spinal cord.
- Ventral root: carries motor messages from the spinal cord to muscles or glands.
- Both roots join to form a spinal nerve, which carries both sensory and motor information.

Functions:

1. Carrying Messages:

The spinal cord acts like a highway of nerves, carrying messages between the brain and the body.

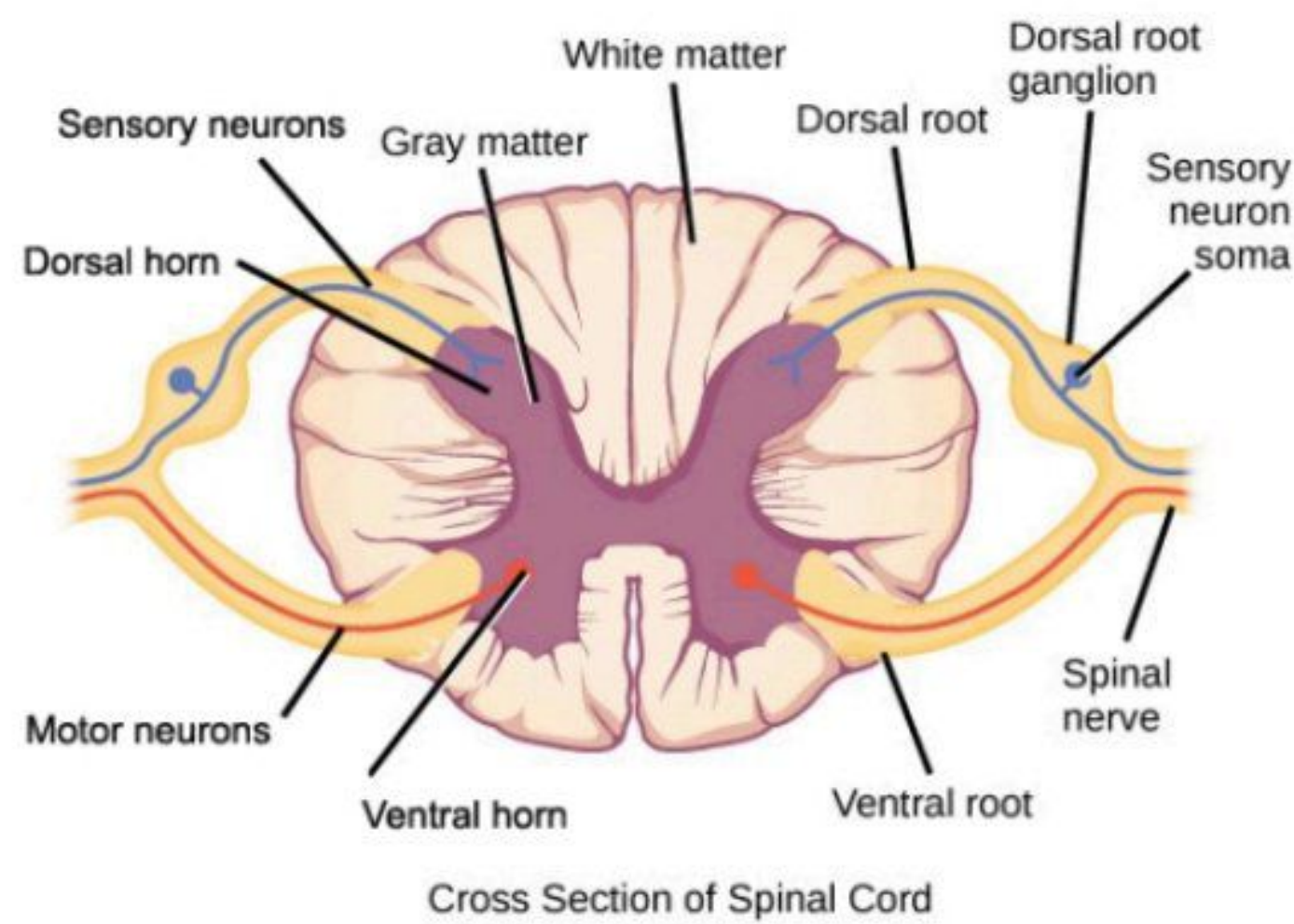
- Sensory nerves carry information to the brain.
- Motor nerves carry instructions from the brain to muscles and glands.

2. Reflex Actions:

The spinal cord controls reflex actions, which are quick, automatic responses to a stimulus. For example, when you touch something hot, you pull your hand back immediately. This action is controlled by the spinal cord, not the brain.

3. Protection and Support:

The spinal cord is protected by the vertebral column and three layers of meninges. The cerebrospinal fluid (CSF) inside it helps cushion and protect it from shocks or injury.

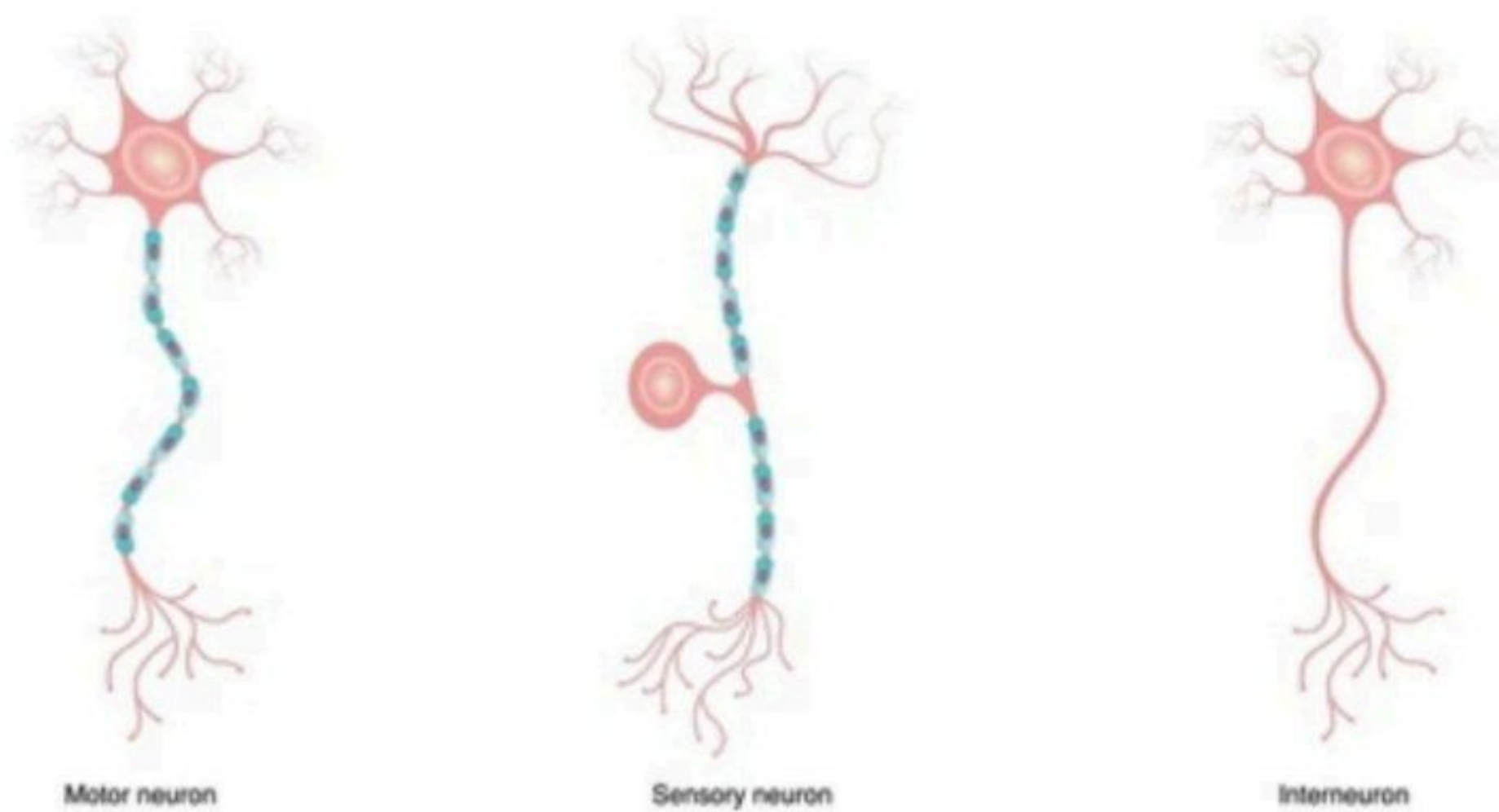


4. Outline the types of neuron with diagram.

Types:

1. **Sensory neuron** – carries impulses from sense organs to CNS.
2. **Motor neuron** – carries impulses from CNS to muscles or glands.
3. **Interneuron** – connects sensory and motor neurons within the CNS.

Types of Neurons



5. Explain that nerve impulses are electrical signals that travel across neurons.

A **nerve impulse** is an electrical signal that travels along the length of a neuron. It is generated when a neuron is stimulated by a change (stimulus). The impulse moves rapidly along the axon in the form of an electrical current. When it reaches the end of a neuron, it is passed to the next neuron through a synapse with the help of neurotransmitters. This process ensures quick communication throughout the body.

6. What are neurotransmitters?

Neurotransmitters are chemical messengers that transmit nerve impulses across the synapse from one neuron to another. Examples include **acetylcholine**, **dopamine**, **serotonin**, and **adrenaline**. They are released from the end of one neuron and bind to receptors on the next neuron, causing a new impulse to be generated. They play a key role in muscle movement, mood, and brain functions.

7. Explain reflex action with diagram.

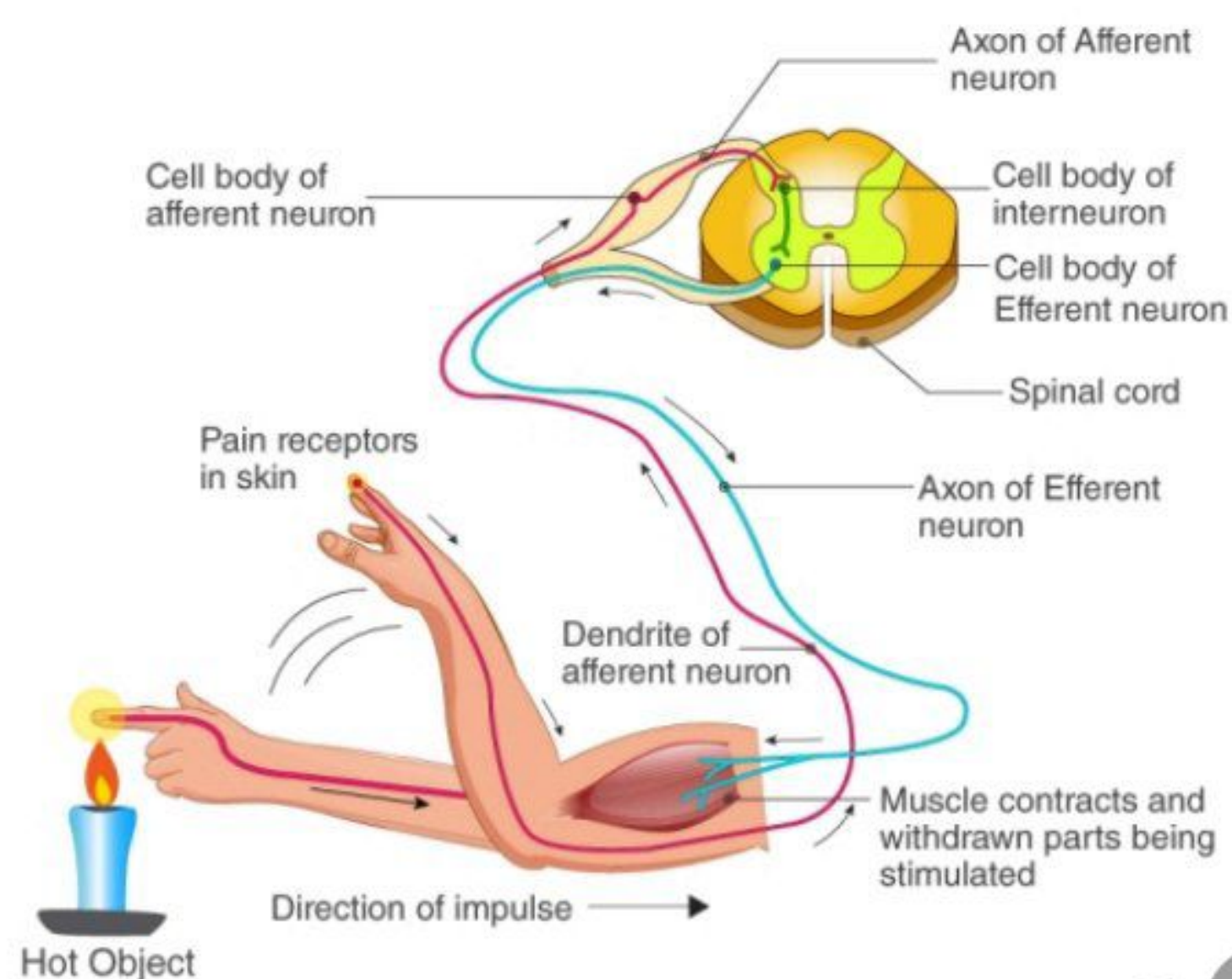
A **reflex action** is a quick and automatic response to a stimulus that protects the body from danger. It involves a **reflex arc**, which includes a sensory neuron, interneuron, and motor neuron.

Example: When you touch something hot, sensory neurons carry the message to the spinal cord, which immediately sends a signal through motor neurons to pull your hand away.

(Diagram: Reflex arc showing path from receptor → sensory neuron → spinal cord → motor neuron → effector.)

REFLEX ACTION

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8. Identify the major endocrine hormones with their functions.

Hormone	Gland	Function
Growth Hormone	Pituitary	Controls body growth and cell development
Thyroxine	Thyroid	Regulates metabolism and energy production
Insulin	Pancreas	Lowers blood sugar level
Adrenaline	Adrenal gland	Increases heartbeat and prepares body for emergency
Testosterone	Testes	Controls male sexual characteristics
Estrogen & Progesterone	Ovaries	Control female reproductive cycle and pregnancy

The endocrine system works along with the nervous system to maintain internal balance and proper functioning of the body.